

Composite capacitor

X16 (2016) — English translation

Capacitor charge

Between plates 1 and 3 of a flat capacitor, a thin metal plate 2 with a charge Q was placed parallel to the plates of the capacitor (Fig. 1). A capacitor is connected to a battery with an emf equal to U_0 . The areas of all three plates are the same and equal to S , and the distance between them is much less than $\sqrt{S}$.

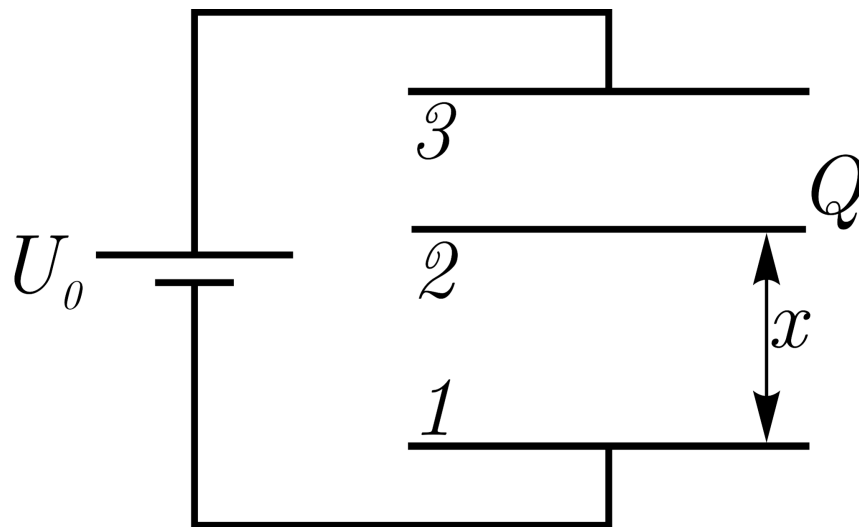


Figure 1: Figure 1.

A1	Find the charges on plates 1 and 3 of the capacitor depending on the distance x between plates 1 and 2.	1.00
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A2	How much work A must be done to move plate 2 from plate 1 to plate 3?	1.00
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Plate 2 was discharged, and the volumes between the plates were filled with dielectric liquids (Fig. 2) with the same dielectric constant ε but with different resistivities ρ_1 and ρ_2 ($\rho_1 < \rho_2$).

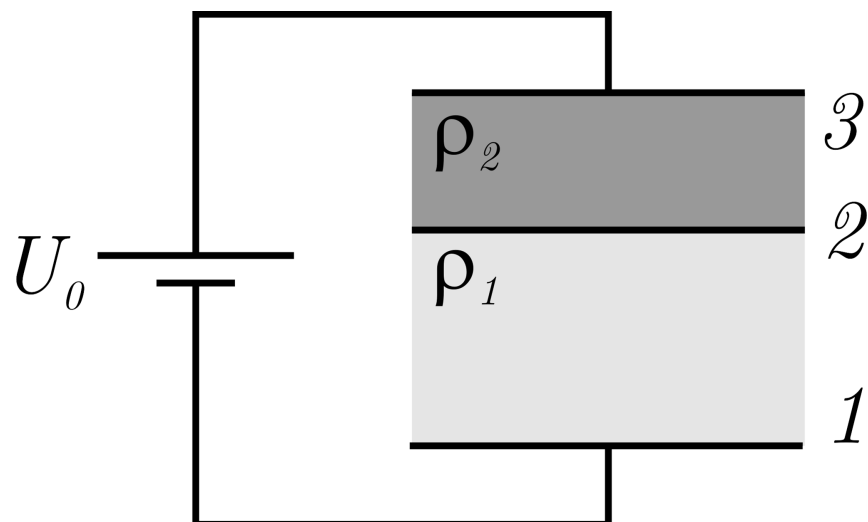


Figure 2: Figure 2.

A3 Find the magnitude and direction of the force acting on plate 2 from the electric field.	3.00
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Source: <https://pho.rs/p/4004>